# SMART RECRUITMENT

# ASSISTANT

AI-Powered Candidate Screening & Ranking

Graduation Project


**TEAM MEMBERS**

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# Problem

### 01 Training Investment

Large numbers of candidate profiles can make initial screening time-consuming.

### 02 MANUAL WORK

Manual review may require significant effort and consistent criteria.

### 03 PRIORITY

Recruiters need a way to prioritize candidates for further review.

### 04 DATA

Use machine learning to provide a data-driven screening signal.

**Goal: Support recruiters — not replace human decisions.**

# Dataset



## HR Analytics: Job Change of Data Scientists

**20,000+**
Training records

**14**
Original features

**8000+**
Null rows detected

**3.01**
Approx. class imbalance ratio

### Examples of features

Education • Experience • Company Information • City • Training Hours • Gender

**Target: Job Change / No Job Change**

# Approach

01

EDA

02

Data Cleaning

03

Feature Encoding

04

Model Training

05

Evaluation

06

Candidate Ranking

## Models

**Logistic Regression • Random Forest • XGBoost**

Stratified 5-Fold Cross Validation + Hyperparameter Tuning

# Dataset Preprocessing



## 01 Data Cleaning & Feature Transformation

### Missing Value Imputation

- We Filled missing numerical fields using the **Median** and missing categorical values with the label **"Unknown"**.
- Machine learning algorithms cannot handle null/blank cells.
- The *Median* was selected because it is robust against skewed distributions and extreme outliers (such as high training hours).
- The *"Unknown"* label treats missing categorical information as a distinct, potentially meaningful behavioral signal.

## 02 Categorical Encoding

- Converted text attributes into numerical values using **Ordinal Mapping** (e.g., Education Level, Experience) and **One-Hot Encoding** (e.g., Gender, Major Discipline).
- *Ordinal Encoding* preserves the natural ranking and hierarchy between levels (e.g., Ph.D. > Masters > Graduate).
- *One-Hot Encoding* creates independent binary columns to prevent models from assuming a false numeric order among nominal categories

# Scaling, Selection & Pipeline Integration

## 01 Data Cleaning & Feature Transformation

### Feature Standardization

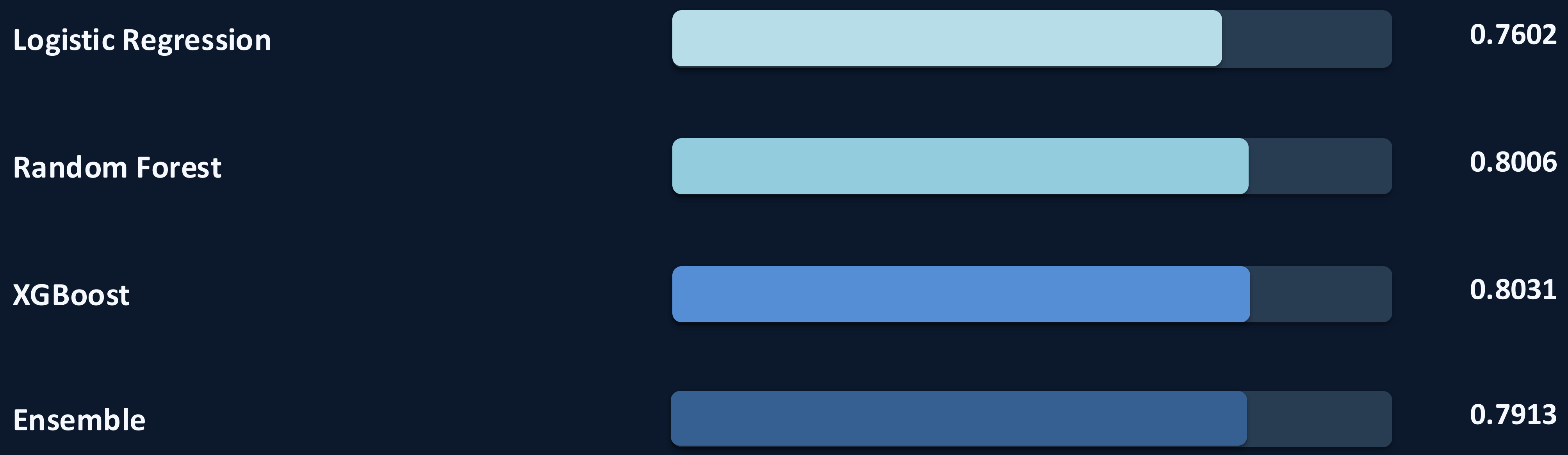
- Applied **StandardScaler** to all numerical inputs to normalize their mean to 0 and variance to 1
- Variables existed on vastly different scales (e.g., City Development Index from 0 to 1 vs. Training Hours up to 300+). Scaling prevents high-magnitude features from disproportionately dominating model weights.

## 02 Feature Selection & Pipeline Encapsulation

- Used **SelectFromModel** with a tree-based estimator to drop the bottom 50% least informative features, bundling all preprocessing and model steps into a **scikit-learn Pipeline**.
- *Feature Selection* eliminates noise created by wide one-hot expansions, preventing overfitting and speeding up convergence.
- *The Pipeline* guarantees strict **Data Leakage Prevention** by ensuring transformations are learned solely within training folds during cross-validation

# Results

## ROC-AUC Comparison




**RECALL**

0.827

**F1-SCORE**

0.865

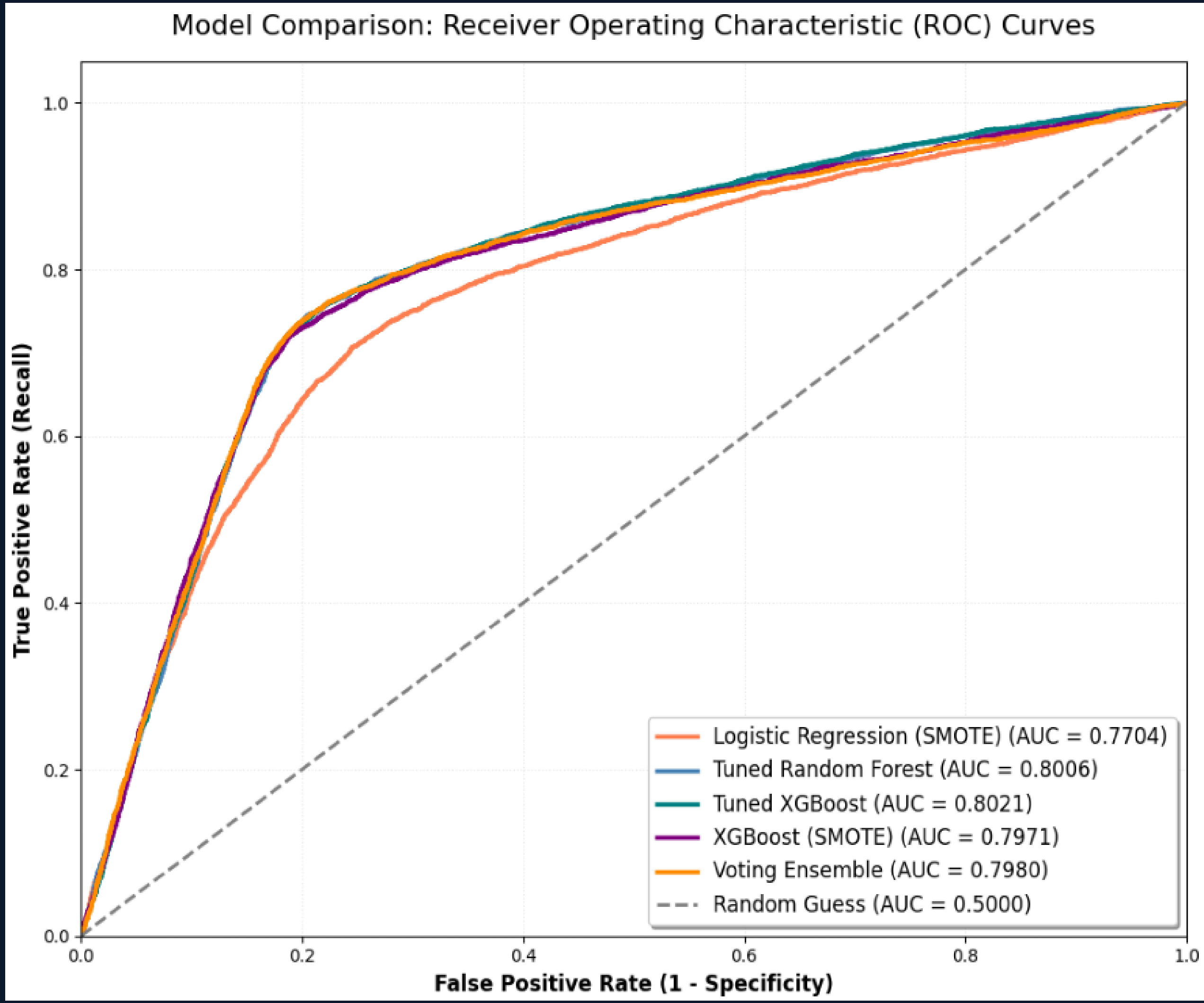
**BEST MODEL**

Ensemble

# Results



## ROC-AUC Comparison



# Business Impact



### SAVE TIME

Reduce effort during initial candidate screening.

### PRIORITIZE

Rank candidates using model probability scores.

### DATA-DRIVEN

Provide a measurable screening signal.

### HUMAN IN LOOP

Support recruiters; final decisions remain human.

# Conclusion



## End-to-End Machine Learning Recruitment Workflow

**Data → EDA → Preprocessing → Modeling → Best Model: Ensemble → Evaluation → Candidate Ranking**

**BEST ROC-AUC**

0.8003

**BEST RECALL**

0.8265

**BEST F1**

0.8638

## Thank You